

# A lender portal, an admin console, and a report service worth paying for

|             |                           |
|-------------|---------------------------|
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| FOR         | The founder               |
| DATE        | 17 September 2026         |
| FOLLOWS     | Launch brief, 16 Sep 2026 |
| STATUS      | Plan, for agreement       |

This covers where each piece runs, what a lender can do after signing in, what the admin console controls, how the reports change against Metropol's own PDF, and what the developer documentation promises. Section 9 lists your jobs and section 10 lists mine, so we can work at the same time.

THE PLAN ON ONE PAGE

|                                                                                                                                                                                                                 |                                                                                                                                                                                    |                                                                                                                                                                                     |                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>ENGINE</p> <p><b>servicesuite-in</b></p> <p>Registry API, exposure broker, report renderer and statement engine, on the same machine as the Postgres. The database is never reachable from the internet.</p> | <p>FACE</p> <p><b>Vercel</b></p> <p>interchange.servicesuitecloud.com: landing page, docs, preview gallery, lender portal and admin UI. API calls are forwarded to the engine.</p> | <p>THIS SESSION</p> <p><b>Preview live</b></p> <p>Landing page, developer docs, the report gallery on an anonymised copy of Daniel's file, and the crunch theatre in demo mode.</p> | <p>BLOCKED ON YOU</p> <p><b>5 things</b></p> <p>Logo pack, database and roles, Funnel port, Vercel access, and Metropol's permission to resell.</p> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

What a lender can do once signed in

| # | CAPABILITY          | WHAT THEY GET                                                                                                                                                                                                                | STATE TODAY           |
|---|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 1 | Bureau Direct       | All 13 Metropol report types on Micromart's contract. Each pull comes back three ways: a PDF on the Interchange letterhead, JSON with every code translated into words, and Metropol's exact response with its SHA-256 hash. | ENGINE BUILT, OFF     |
| 2 | Interchange reports | The nine-product catalogue from the launch brief: identity, delinquency, score, affordability, credit file, exposure, intent, contactability, cohort.                                                                        | 4 LIVE<br>5 SPECIFIED |
| 3 | Statement Crunch    | Upload an M-PESA PDF and its password, and the same crunching sequence as Micro Eazy plays. Crunch as many statements as needed and download each result.                                                                    | TO PORT               |
| 4 | Live exposure       | Query every member's book at once, with a radar view that lights up each member as it answers and shows its real latency.                                                                                                    | ENGINE LIVE           |
| 5 | Workspace           | Enquiry history and re-downloads, consent ledger, team, API keys and sandbox, usage and invoices, contribution health.                                                                                                       | TO BUILD              |

What the admin console governs

For every member it shows the configuration and the monthly usage: statements crunched, reports pulled by type, bureau pulls and spend against the cap, and exposure queries asked and answered. It also shows positions contributed, how fresh the book is, a reciprocity score, answer rate and p95, and refusals by reason. From the same console you promote or suspend members, set caps and prices, manage keys and operators, switch the bureau on or off, and export invoices and audit evidence.

THE ONE THING THAT CAN STOP THE BUREAU HALF

A disclaimer covers how the data is presented. It does not cover breaking the subscription contract. Before any member other than Micromart pulls a Metropol report, we need Metropol's written confirmation that reports may be pulled through the Interchange, and under whose subscription. Until then, Bureau Direct is on for Micromart only, and Axe gets the ecosystem products. The wording to send Ambale is in section 9, item E1.

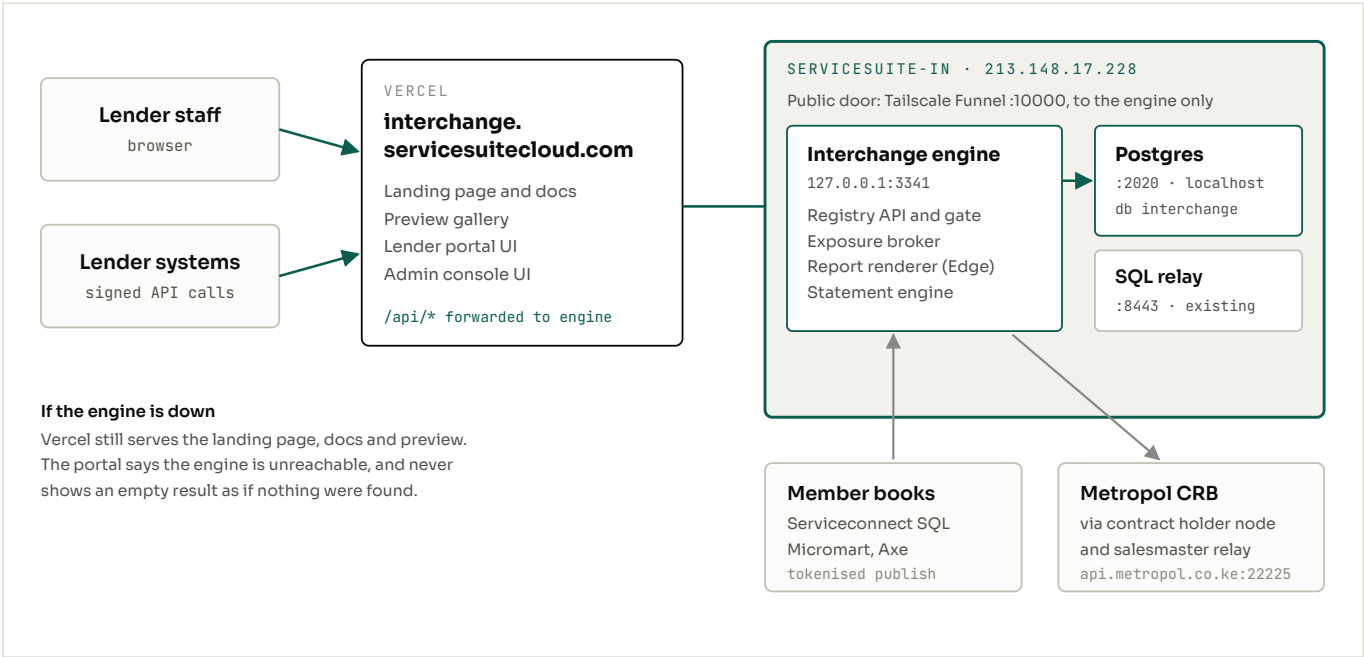
# What I checked before writing this

Every decision below rests on one of these checks. Where something could not be checked, the table says so.

| CHECK                          | RESULT                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What servicesuite-in is        | Tailnet address 100.115.241.36 , public address 213.148.17.228 . This is the same machine whose Postgres CallBox connects to ( C:\GIT\CallBox\CallBox\appsettings.json , host 213.148.17.228 , port 2020 , databases FastDialCallBoxMaster and serviceconnect ).                                                                                                                                                                   |
| Postgres port 2020             | Open on the tailnet (29 ms) and on the public IP (40 ms) from this workstation. Port 5432 is closed. Ports 443 and 8443 answer on the tailnet.                                                                                                                                                                                                                                                                                     |
| SQL relay                      | /health on :8443 returns ok, writes true, 39 served, 0 refused, up since 16 Sep 01:37 UTC. You said the node is back, and it is.                                                                                                                                                                                                                                                                                                   |
| Postgres internals             | Not read. A read-only probe (version, TLS, roles, databases) was blocked by this session's permission layer, so I did not try another way round it. Item B3 asks you for the four values instead.                                                                                                                                                                                                                                  |
| Whose file "Daniel" is         | National ID 30058967, the Micromart Fintech borrower pulled on 15 Sep. We already hold Metropol's exact answers for reports 3, 8, 11, 12 and 16, plus Metropol's own PDF ( XM8B-YR06-20260915 ). The preview needs no new pull for those five.                                                                                                                                                                                     |
| Metropol API guide v3.8        | Read in full (53 pages). It defines 14 report types, 4 report reasons, 5 identity types, 5 delinquency codes, 5 lender sectors, 15 product types, 18 account statuses, 34 response codes and a health endpoint. The guide says it is proprietary and may not be reproduced without Metropol's written approval. See section 6.                                                                                                     |
| Metropol's PDF against its API | For the same person on the same afternoon, the portal PDF lists 50 accounts (46 bank, 4 digital credit provider). API report 8 lists 48 (44 bank, 4 other). The total outstanding matches at 39,106.94. The PDF also prints a probability of default (67%), which no JSON report returns. So our documents must say which source each figure came from, and a PD tile can only appear when report 4 (the PDF) is part of the pull. |
| Entitlement                    | 13 of 14 report types are entitled, per the free probe on 16 Sep. Report 22 is refused with E029.                                                                                                                                                                                                                                                                                                                                  |
| Vercel                         | This workstation has no Vercel login, and the Interchange repo is not linked to a Vercel project. Item D1 fixes that.                                                                                                                                                                                                                                                                                                              |
| The console as built           | Operator roles already exist: SUPER_ADMIN sits above all members, and MEMBER_ADMIN, ANALYST and AUDITOR are each scoped to one member. Also built: a rights catalogue checked in proxy.ts , signed sessions, lockout, the signed POST /api/v1/report in Metropol's request shape, a hash-chained log, and governance actions. The report kit is in lib/reports/ . About 25 files from 16 Sep are still uncommitted.                |

# Where it runs: the engine sits beside the database, the face runs on Vercel

You asked for the important engines on the main SQL relay host. That is also the only arrangement that meets the latency target without putting a database on the internet.



### Why not all on Vercel

Vercel cannot join the tailnet, and on standard plans it has no fixed outbound IPs. The database would have to accept connections from the whole internet. That is not a trade the most secure resource in the estate should make.

### Why this fixes latency

The brief measured p95 at 1,182 ms, and nearly all of that was round trips to a Postgres in Ireland. The fan-out itself took about a millisecond. With the engine beside Postgres, every member read happens on the same machine, so the 400 ms target becomes a real test rather than a distance.

### Why the engine renders PDFs

The report kit prints through a headless Chromium. The Windows host can run Edge. On Vercel we would have to bundle a Chromium and work around its size limits, for the documents that matter most.

## Guard rails for a shared machine

This machine already runs CallBox, the PBX link and the SQL relay. The Registry is a tenant there, just as Micro Eazy is a tenant on Micromart's server. It gets its own roles, a connection limit, statement timeouts and a memory ceiling on the engine process. Book ingest runs off-peak. I never restart Postgres: `pg_hba` changes are applied with `pg_reload_conf()`, which leaves CallBox's connections alone.

## The lender portal, capability by capability

### 1 Bureau Direct: the Metropol reports, packaged properly

| TYPE | REPORT                        | WHAT THE LENDER READS                                               | PREVIEW SAMPLE |
|------|-------------------------------|---------------------------------------------------------------------|----------------|
| 1    | Identity Verification         | Is this ID real: name, date of birth, gender, serial, deceased flag | SECTION OF 12  |
| 2    | Delinquency Status            | Codes 001 to 005 in plain words, with the loan amount asked about   | SECTION OF 12  |
| 3    | Metro / Mobile Score          | The score, where it sits in the range, and when it was calculated   | CAPTURED       |
| 4    | PDF Credit Report             | Metropol's own document, handed over unchanged                      | NOT SHOWN      |
| 5    | JSON Credit Report            | Accounts, sectors, enquiries, PPI, reported names                   | NEEDS 1 PULL   |
| 6    | Identity Scrub                | Phones, emails, addresses, employers, and how the names vary        | SECTION OF 12  |
| 8    | Credit Info                   | The account ladder, sector split, enquiry windows, score            | CAPTURED       |
| 10   | Enhanced Credit Info          | Report 8 plus guarantors and stakeholders                           | NEEDS 1 PULL   |
| 11   | Credit Info + Income Estimate | Estimated income against the instalments already committed          | CAPTURED       |
| 12   | Full Enhanced Credit Info     | Everything, with the 12-month score and PPI trend                   | CAPTURED       |
| 13   | Minified Credit Info          | One line: arrears days, NPL flag, outstanding balance               | NEEDS 1 PULL   |
| 14   | Full JSON Report              | Identity, scrub, credit info, guarantors, stakeholders              | NEEDS 1 PULL   |
| 16   | Credit Accounts Summary       | Mobile against generic loans, arrears, NPL, monthly instalment      | CAPTURED       |
| 22   | Accounts Info (12-month)      | Not in Micromart's contract (E029). Shown as "request access".      | REFUSED        |

#### Every pull, four downloads

- PDF on the Interchange letterhead (section 5)
- JSON in the Interchange format, with every code turned into words
- Bureau response: exactly what Metropol sent, byte for byte, with its SHA-256
- CSV of the account ladder, for reports that have accounts

For type 4, Metropol's own PDF is passed through untouched.

#### Guard rails before money is spent

- Consent reference and report reason (1 to 4) are required
- The price shows before the pull is confirmed
- Daily cap per member, and a global switch to turn the bureau off
- A repeat of the same ID and report within the member's window returns their stored copy at no charge. It is never served to a different member.
- Metropol's 60-second duplicate rule is explained rather than shown as a bare error

### 2 Interchange reports

This is the catalogue from the launch brief. Types 1, 2, 12 and 20 already answer. I build the rest in order of what unlocks the most: **11 Cashflow & Affordability** first, because Statement Crunch feeds it directly, then **3 Interchange Score**, **23 Cohort Benchmark**, **22 Contactability** and **21 Intent Signal**. Each is the same package as Bureau Direct (PDF, JSON, CSV), and each labels whether a figure came from member books, from the bureau, or was worked out by the Interchange.

### 3

#### Statement Crunch, with the Micro Eazy theatre

The lender uploads an M-PESA PDF and types the password from Safaricom's SMS. The full-screen theatre then plays the same stages in the same order as the app: **decrypt, parse, extract, post to ledgers, audit, score**. Only the receipt codes that flicker while the server works are decoration. From the extract stage on, every figure comes from the actual statement.

##### What comes out

- Monthly income, spend and net
- Spend by life category, and the top merchants
- Other lenders visible in the statement, with amounts borrowed and repaid
- Gambling share
- Affordability and a suggested ceiling on the monthly instalment
- Thin-file score with reason codes

##### Many at once

Several statements can be queued. Each gets a row with its own progress, the theatre plays for the one in focus, and each result downloads as a PDF on the letterhead, JSON (features and transactions) and a CSV of transactions.

A name guard compares the statement holder with the name on the enquiry. A mismatch stops the crunch and says why.

##### What is never kept

The PDF and its password are processed in memory and dropped. The Registry keeps only the derived figures, a digest of the transactions and the file's SHA-256. That is enough to audit, to catch a duplicate, and to bill.

The engine is the one already in connected-suite ( `lib/statement/*` , `pdfjs-dist` ).

### 4

#### Live exposure

An officer enters an ID and a consent. The engine turns the ID into a token and queries every member at once. On screen, a radar sweep shows each member node lighting up as it answers, with that member's real latency. A member that stays silent is named with its reason: timed out, not yet published, or errored. That keeps "partial" from ever being mistaken for "clean". The result is aggregates only, as the exposure policy already sets: how many lenders hold the borrower, the outstanding band, the arrears state and how recently they borrowed.

### 5

#### The rest of the workspace

- **New enquiry:** one ID, tick the products, get one decision pack and pay once
- **History:** every pull and crunch can be downloaded again during the retention period
- **Consent ledger:** captured, used, revoked, and by whom
- **Team:** a member admin invites analysts and auditors
- **API keys and sandbox:** key pair made in the browser, sandbox IDs that cost nothing
- **Usage and invoices:** this month's pulls, crunches and fees
- **Contribution health:** last publish, positions, freshness, reciprocity score
- **Alerts:** bureau down, cap reached, book going stale

## The admin console: oversight of every member

Only SUPER\_ADMIN can open it. Every action taken here goes into the governance record with who decided it and why, and every member can see that record.

### Measured for each member, each month

| MEASURE                     | WHERE IT COMES FROM                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------------------|
| Statements crunched         | New usage events, written by the statement engine for each crunch, including failed and refused ones |
| Reports by type             | Audit entries per service, split into Bureau Direct and Interchange reports                          |
| Bureau pulls and spend      | Metropol's cost passed through, plus the Interchange fee, against the member's daily cap             |
| Exposure queries asked      | Audit entries for the exposure service where this member is the caller                               |
| Queries answered for others | The fan-out records where this member's node answered                                                |
| Contribution                | Positions published, publish generation, and hours since the last publish                            |
| Node quality                | Answer rate, p95 latency, and silent answers by reason                                               |
| Refusals                    | No consent, wrong scope, reciprocity, over quota                                                     |
| Access                      | Active operators, sign-ins, API calls against portal use                                             |

### Reciprocity score, proposed

#### Give against take

**Give** is the member's share of all positions in the network, multiplied by freshness. Freshness is 1.0 when the book was published within 24 hours and falls to zero at 7 days.

**Take** is the member's share of all queries in the month.

**Score** is give divided by take. At 1.0 or above the member is a net contributor. Between 0.5 and 1.0 they are balanced. Below 0.5 they are a net consumer and get a warning.

#### The hard rule stays in code

No fresh publish in 7 days means queries are refused as `REFUSED_RECIPROCITY`, which the gate already enforces. The score is what the admin and the member see before that point. The thresholds above are my proposal, and you set the real ones.

### Controls

- Promote from shadow, suspend, reinstate, each with a written reason
- Per member: products allowed, daily bureau cap, free tier, fee overrides
- API: authentication profile, IP allowlist, webhook address
- Keys: register, rotate, revoke
- Operators: create, disable, reset code, require a passkey
- Bureau: one global switch plus a switch per member
- Catalogue and tariff: type in Metropol's rate card once and every quote re-prices
- Invoices: a monthly PDF per member, bureau cost and Interchange fees on separate lines
- Audit export, with the hash chain re-verified in the file
- Health: engine, database, relay, bureau, and member node latency

## The report, measured against Metropol's own PDF

Compared: Metropol's standard credit report ( `XM8B-YR06-20260915` , 7 pages) and our Credit File from 16 Sep. Their document is a complete record with no analysis. Ours has the analysis, but it lacks the letterhead, the regulatory text and the request details that make a document look official.

| ELEMENT                 | METROPOL PDF                                          | OUR CREDIT FILE TODAY                                    | TARGET                                                                                                               |
|-------------------------|-------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Letterhead              | Logo, with address, phone, email and web on the right | Wordmark in text only                                    | Your logo, the operating company's address block, and the document reference                                         |
| Regulation 40(1) notice | Boxed, top of page 1                                  | Missing                                                  | Boxed, top of page 1, word for word, credited to the regulation                                                      |
| Request details         | Requested by, organisation, date, reference           | Member code, token, generated time                       | Requested by (a person), requesting organisation, report date in EAT, Interchange reference, Metropol transaction ID |
| Headline figures        | Metro-Score, PPI, probability of default              | Score, exposure, overdue, adverse, new credit            | Keep ours and add a PPI tile. Show PD only when report 4 is in the pull, marked as from Metropol's PDF.              |
| Identity                | Verified ticks, latest and older addresses            | Identity grid and traces                                 | Add verified marks and a latest-or-older marker, drawn as SVG                                                        |
| Sector table            | All, NPA, performing with and without default history | Bar chart only                                           | Add the table under the chart                                                                                        |
| Accounts                | Every account, 5 pages                                | Largest balances only                                    | Decision view first, then every account as an appendix with product and status codes in words                        |
| Reading of the file     | None                                                  | Summary, trend, arrears, activity strip, staleness flags | Keep. This is what the lender pays for.                                                                              |
| Closing notice          | Metropol disclaimer                                   | Missing                                                  | Metropol's text word for word and credited, followed by the Interchange notice below                                 |
| Verification            | None                                                  | None                                                     | A QR code and reference that open a page confirming the document's hash against the message log                      |
| Footer                  | Page X of Y                                           | None                                                     | Page X of Y, reference, and a confidentiality line                                                                   |

### Provenance on every block

Each section carries a small source line: `Source: Metropol CRB · report 12 · trx 3f1c...` , `Source: member books · 4 of 4 answered` , or `Interchange reading of bureau data` . A lender can always tell whose statement a figure is, and so can a regulator.

Notices, in the order they print

TOP OF PAGE 1 · WORD FOR WORD

Disclaimer: According to the Banking (Credit Reference Bureau Regulations, 2020 Regulation 40 (1), A customer's credit score shall not be used solely to deny the customer a loan, credit facility, or any other financial service. However, it shall be used as one of the factors to inform the decision-making when determining the customer's application for a loan, credit facility, or any other financial service.

REQUEST ROW · AS ON METROPOL'S REPORT, EXTENDED

Requested By: *operator name* · Requesting Organisation: *member legal name* · Report Date: 2026-09-15 16:07:28 EAT · Reference Number: #REF IX-XXXX-XXXX-20260915 · Bureau Transaction: *Metropol trx\_id*

LAST PAGE · METROPOL'S DISCLAIMER, WORD FOR WORD

DISCLAIMER: The information provided in this report is for the purposes of decision making as stipulated in the Banking (Credit Reference Bureau) Regulations, 2020 only. It is not in any way METROPOL CREDIT REFERENCE BUREAU's opinion of the financial capability, solvency, integrity or intentions of any persons mentioned herein. It therefore cannot be used to inform any level of decision making, positive or adverse, on any and all persons mentioned herein, whether regarding current or potential credit lines. All parties given access to this information are bound to keep any and all information, whether intentionally or accidentally accessed, strictly confidential. METROPOL CREDIT REFERENCE BUREAU does not warrant or guarantee any use of the information contained herein.

LAST PAGE · THE INTERCHANGE NOTICE · DRAFT FOR COUNSEL

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The part in square brackets is your decision (item A2). It is accurate and shows where the data came from, but it names your contract holder on every document a third party reads.

Formats and names

| FILE            | CONTENTS                                                                                       | NAME PATTERN             |
|-----------------|------------------------------------------------------------------------------------------------|--------------------------|
| PDF             | The lettered document above                                                                    | IX-Credit-File<ref>.pdf  |
| JSON            | Interchange envelope: codes turned into words, provenance on each block, response codes, quote | IX-Credit-File<ref>.json |
| Bureau response | Metropol's body exactly as sent, one file per report type, plus its SHA-256                    | IX<ref>-metropol-12.json |
| CSV             | Account ladder, or statement transactions                                                      | IX<ref>-accounts.csv     |

## The developer API and its documentation

A lender already wired to Metropol should be able to change a base URL and a key pair and keep their parsing code. The request fields and the E-codes stay the same. The authentication is stronger.

### The credential pack a member receives

| ITEM                           | VALUE                                                                                                                                                    |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Base URL, port, version</b> | <code>https://interchange.servicesuitecloud.com:443/api/v1/</code> . The sandbox is the same address with <code>/sandbox/v1/</code> .                    |
| <b>Member code</b>             | <code>KE/LENDER/3002</code> (X-Road style, already in the Registry)                                                                                      |
| <b>Public key</b>              | Registered with the Interchange. It identifies the member.                                                                                               |
| <b>Private key</b>             | Generated in the lender's own browser in the portal and downloaded once. The Interchange never sees it, so nobody can sign as that lender, including us. |
| <b>IP allowlist</b>            | The addresses their systems call from. It is required for production keys.                                                                               |
| <b>Webhook secret</b>          | For <code>report.ready</code> , <code>crunch.completed</code> and <code>cap.reached</code>                                                               |

#### Authentication: signed (at launch)

The member signs each request with Ed25519 over the canonical request: method, path, the SHA-256 of the body, timestamp, nonce and member code. This is already built, and requests more than 60 seconds old are refused. Still to add: a nonce store, so that a replay inside that minute is refused as well.

#### Authentication: compatible (later, if asked)

The same headers as Metropol (key, hash, timestamp) for lenders who cannot add a signing library. The hash is HMAC-SHA256 rather than Metropol's plain SHA-256 of joined strings, and an IP allowlist is mandatory. I would not build this until a member asks for it.

### Endpoints

| CALL                                                      | PURPOSE                                                                                                   |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| POST <code>/api/v1/report</code>                          | Any report, Bureau Direct or Interchange. <code>format</code> is json, pdf or verbatim. Built.            |
| POST <code>/api/v1/exposure</code>                        | Live cross-member exposure, with a per-member answer status                                               |
| POST <code>/api/v1/statement/crunch</code>                | Multipart upload: PDF, password, name to check against. Returns the crunch JSON, or a job ID for batches. |
| GET <code>/api/v1/report/{ref}</code>                     | Download a past document again in any format, during the retention period                                 |
| POST <code>/api/v1/consent</code> · <code>/revoke</code>  | Issue or withdraw a consent reference. Built.                                                             |
| GET <code>/api/v1/catalogue</code> · <code>/health</code> | Products, prices and states. Engine, database and bureau status.                                          |

## Code tables

Every code value Metropol returns is passed through **unchanged**, so a lender's switch statements keep working. The Interchange adds the description in its own words, and adds its own error range that never collides with Metropol's.

| TABLE            | VALUES                                                                       | CODE      | INTERCHANGE<br>MEANING                                                                    |
|------------------|------------------------------------------------------------------------------|-----------|-------------------------------------------------------------------------------------------|
| Report types     | 1-6, 8, 10-14, 16, 22                                                        | IX101     | No consent reference                                                                      |
|                  |                                                                              | IX102     | Consent revoked or expired                                                                |
| Report reasons   | 1 new · 2 review · 3 verify · 4 own request                                  | IX103     | Consent does not cover this product                                                       |
|                  |                                                                              | IX104     | Consent belongs to a different borrower                                                   |
| Identity types   | 001 ID · 002 passport · 003 service · 004 alien · 005 company                | IX201     | Member is in the shadow period                                                            |
|                  |                                                                              | IX202     | Member's book is stale; contribute to query                                               |
| Delinquency      | 001 not found · 002 no accounts · 003 clean · 004 current NPA · 005 past NPA | IX301     | Daily bureau cap reached                                                                  |
|                  |                                                                              | IX401     | Bureau unreachable                                                                        |
| Lender sectors   | bank · mfb · mfi · sacco · other                                             | IX402     | Bureau refused; its code is in <code>upstream.api_code</code>                             |
|                  |                                                                              | IX501-504 | Statement: password needed, password wrong, not an M-PESA statement, holder name mismatch |
| Product types    | 1-14, 18, 19 (15-17 unused)                                                  | IX601     | Partial answer: named members did not respond                                             |
| Account status   | A-R, N, and "-" for unknown                                                  |           |                                                                                           |
| Bureau responses | E001-E033, E200, E409                                                        |           |                                                                                           |

## What the docs also explain, learned from live pulls

- `api_code` sometimes arrives as a number and sometimes as a string. The Interchange always returns a string.
- E017 on an entitled report is a thin file, not a failure.
- Arrears days count to the lender's last upload ( `loaded_at` ), not to today. The Interchange adds the age of that figure.
- Bureau enquiry counters are not real-time. Four pulls did not move them within 29 minutes.
- Metropol's PDF and its API can list different account counts for the same person on the same day.

METROPOL'S GUIDE IS PROPRIETARY

The guide states it "shall not be reproduced or disclosed in whole or in part" without Metropol's written approval. Our docs describe **our** API in **our** words, and they publish code values that appear in every response anyway. We do not republish the guide, its examples or its tables. Item E1 asks Metropol to confirm this is acceptable.

## The docs site

The docs live at `/docs` on Vercel and hold:

- A five-minute quickstart: make keys in the portal, sign a request, make the first sandbox call.
- A reference page for every endpoint, with request and response samples from the anonymised file.
- The code tables and an error handbook.
- Snippets in Node, C#, PHP and Python. C# matters because ServiceSuite lenders run .NET.
- A Postman collection, a changelog and a status page.

## Security: what "most secure" means in practice

### Already built

- Every member call signed with Ed25519
- OPRF tokens, so no raw ID reaches the Registry
- A hash-chained message log with a tamper test
- Signed sessions checked in `proxy.ts`, and one rights catalogue
- Lockout on access codes
- TLS required for any database not on the same machine
- The bureau key never held by the Registry
- Consent tied to the borrower's token

### Added in this build

- Database reachable only from the same machine and the tailnet
- App role without BYPASSRLS, proven with a test (brief action 9)
- Passkey sign-in for admins, with TOTP as the fallback
- Per-member IP allowlist, and rate limits per member and per ID
- Statement files and passwords never stored
- A QR code on every document that verifies its hash
- CSP and HSTS headers, and no framing
- Nightly encrypted backup of the Registry

FOUND TODAY, AND IT MATTERS BEFORE THE REGISTRY MOVES

`C:\GIT\CallBox\CallBox\appsettings.json` holds the **Postgres superuser password** in plain text, for port 2020, which **answers from the public internet**. Anyone who has that repository can connect as superuser and read every database on that server, and that will include the Registry once it moves there. A superuser skips every role and row-level rule we add.

My recommendation: rotate that password, take it out of the repository, and limit port 2020 to the addresses that need it. This touches CallBox, so time it with whoever runs CallBox. The Registry can move before this is done, but until it is, its protection is only as strong as that password.

### Still open from the launch brief

- **OPRF key custody.** The key is still in an `.env` file. Moving it to a KMS or HSM is cheap now and very expensive once more members' tokens depend on it (brief action 8).
- **Consent from borrowers themselves.** Today the console mints consent. Counsel's condition is that consent can be proved at every hop (brief action 7).
- **RFC 3161 timestamps on the log.** The chain proves the order of entries, but not the wall-clock time they were written.

## The preview on Vercel, built on Daniel's file

The preview shows exactly what a live request returns, using real bureau answers with the identity replaced. Nothing is invented. Where we have no captured answer, the gallery says so rather than making one up.

### Pages

- `/` landing page with the network in motion, drawn from real totals: 17 members, 75,025 positions, 2,252 held by two or more members
- `/docs` developer documentation
- `/preview` gallery: every product as a web view, PDF, JSON, bureau response and CSV
- `/preview/crunch` the theatre on a sample statement
- `/preview/exposure` the radar, replaying the 16 Sep measurement and labelled as a replay

### How the file is anonymised

- Replaced:** name and every reported spelling; the ID (becomes a number from Metropol's test set, which exists on no real file); phone; date of birth; addresses; ID serial; transaction IDs; the Metropol reference
- Account numbers are rewritten.** 42 of them contain his phone number ( 2547...2020... ).
- Kept:** amounts, dates, product types, statuses, arrears, sector counts, score, PPI. The API data carries no lender names.
- Every page is marked **SAMPLE · ANONYMISED**.
- Leak scanner:** the build fails if any generated file contains the real ID, phone, name parts, serial or phone-derived account fragments.

### Which samples are real

| REPORT TYPES     | SAMPLE IN THE PREVIEW                                                                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3, 8, 11, 12, 16 | Metropol's captured answers from 15 Sep, anonymised. <span>REAL</span>                                                                                                                                                 |
| 1, 2, 6          | These sections exist inside the report 12 answer, so they are shown as that and labelled "section of the report 12 answer". A standalone call returns the same fields in its own envelope. <span>REAL, LABELLED</span> |
| 5, 10, 13, 14    | No capture exists. Either you approve one billed pull of each on 30058967 (item F3), or the gallery shows the field layout with no values. <span>YOUR CALL</span>                                                      |
| 4                | Metropol's own document. Anonymising it would mean altering the bureau's PDF, so the gallery describes it and shows no sample. <span>DESCRIBED</span>                                                                  |
| Statement Crunch | A real statement you provide (item F2), anonymised the same way. Until then, a synthetic sample labelled as synthetic. <span>PENDING F2</span>                                                                         |

## Your side

These are grouped so you can hand parts to other people. Each item ends with what to send back. B, C and D unlock the live engine. A and F make the preview look and read like the real thing.

### A Brand and visuals

---

#### A1 The Interchange logo pack

You need five files:

- An SVG master of the horizontal lockup
- An SVG of the mark alone, square, that still reads at 16 px
- A white version for dark screens
- A single-colour black version for print
- PNG copies at 2048 px with transparent backgrounds

If brand colours are defined, send their hex values. If not, I take them from the logo.

**SEND** Drop the files in `interchange\apps\interchange-console\public\brand\`

---

#### A2 Letterhead facts

I need:

- The legal name of the company that operates the Interchange
- Physical address, P.O. Box, phone, email and website
- Company registration number
- ODPC registration number, if registered
- Whether the notice should name Micromart as the bureau subscriber (the square brackets in section 5)

**SEND** A short reply with the details

---

#### A3 Photos and the hero video

The list with specs is in section 11. Only the logo is required. Everything else has a stand-in already in the repo.

**SEND** Files into `public\media\` with the names in section 11

---

#### A4 Safaricom marks in a paid product

The Micro Eazy theatre uses Safaricom's logo animation and the M-PESA backdrop. That is one thing inside a customer app and another inside a product we sell to lenders. You have two options:

- Port it exactly as it is in the app.
- Keep the green, the ring and the sequence, and leave out Safaricom's logo.

**SEND** "as is" or "without the logo"

---

## B1 Create the roles and the database

Run this as `postgres` on that machine, in `psql` or `pgAdmin`. Replace each password with a new random one of at least 32 characters. It creates objects only: nothing restarts, and `CallBox` is not touched.

```
-- Hash new passwords with SCRAM (skip this line if the server is older than Postgres 10)
SET password_encryption = 'scram-sha-256';

-- Two roles: neither is a superuser, and neither bypasses row-level security
CREATE ROLE interchange_owner LOGIN PASSWORD '<new password 1>'
    NOSUPERUSER NOCREATEDB NOCREATEROLE NOBYPASSRLS CONNECTION LIMIT 5;
CREATE ROLE interchange_app LOGIN PASSWORD '<new password 2>'
    NOSUPERUSER NOCREATEDB NOCREATEROLE NOBYPASSRLS CONNECTION LIMIT 40;

-- The database, and nobody else may connect to it
CREATE DATABASE interchange OWNER interchange_owner ENCODING 'UTF8' TEMPLATE template0;
REVOKE ALL ON DATABASE interchange FROM PUBLIC;
GRANT CONNECT, TEMPORARY ON DATABASE interchange TO interchange_app;

-- Limits so the Registry can never hold CallBox's server up
ALTER ROLE interchange_app SET statement_timeout = '15s';
ALTER ROLE interchange_app SET idle_in_transaction_session_timeout = '30s';
ALTER ROLE interchange_owner SET statement_timeout = '10min';

-- Inside the new database: only its owner may create objects
\c interchange
REVOKE CREATE ON SCHEMA public FROM PUBLIC;
ALTER SCHEMA public OWNER TO interchange_owner;
```

I then create the tables with `registry:bootstrap --apply --grant interchange_app`, which gives the app role only the table rights it needs.

## B2 Only the machine itself and the tailnet may reach it

Open the file that `SHOW hba_file;` names. Add these lines **above** any existing line that allows `0.0.0.0/0`, because the first matching line wins. Then run `SELECT pg_reload_conf();`. A reload applies the change without dropping anyone's connection.

```
# The Interchange Registry: this machine and the tailnet only
host interchange interchange_owner,interchange_app 127.0.0.1/32 scram-sha-256
host interchange interchange_owner,interchange_app ::1/128 scram-sha-256
host interchange interchange_owner,interchange_app 100.64.0.0/10 scram-sha-256
host interchange all 0.0.0.0/0 reject
host interchange all ::/0 reject
```

## B3 Tell me what the server is, and hand over the connection strings

```
SELECT version(); SHOW ssl; SHOW password_encryption; SHOW hba_file;
```

Put the two URLs in `interchange\apps\interchange-console\.env`. Do not paste them in chat. My workstation reaches the server over the tailnet address. On the machine itself, the engine uses 127.0.0.1.

```
REGISTRY_ENGINE_DATABASE_URL="postgresql://interchange_app:<pw2>@100.115.241.36:2020/interchange"
REGISTRY_ENGINE_DIRECT_URL="postgresql://interchange_owner:<pw1>@100.115.241.36:2020/interchange"
```

**SEND** The four values from the query. If `ssl` is off, I treat the tailnet's WireGuard encryption as the secure link, and nothing ever connects over the public IP.

## C The engine host (servicesuite-in)

### C1 What the machine has

I need:

- Windows version
- Free RAM and free disk space
- The output of `node -v` (the relay already runs Node)
- Whether Microsoft Edge or Chrome is installed, which the PDF renderer needs

**SEND** These in a short reply

### C2 A home for the engine

Create `C:\Interchange\`. I will give you a one-time install script that places the build, installs it as a Windows service and writes logs to `C:\Interchange\logs`. You run it once, and every later update is a single command.

**SEND** "Folder ready"

### C3 The public door, and only for the engine

Run `tailscale funnel status` and send me the output. It shows what already uses 443 and 8443. Funnel only allows ports 443, 8443 and 10000, and 8443 is the SQL relay, so the engine will most likely use 10000. Once the engine is running, I will give you the exact funnel command for your Tailscale version. Do not open port 3341 in the Windows firewall, because Funnel handles TLS.

**SEND** The funnel status output, and the Tailscale version

## D Vercel and the domain

### D1 Let me deploy

**Recommended:** create a Vercel token under Account Settings, Tokens, scoped to the right team, and put it in

`interchange\apps\interchange-console\.env` as `VERCEL_TOKEN=`. I then deploy and iterate from here, and nothing needs pushing to GitHub first.

**The other option:** import `emmanuel-arch/Interchange` in Vercel with Root Directory `apps/interchange-console`, and let me push.

**SEND** "Token in .env" plus the team name, or "Repo imported"

### D2 The domain

Add `interchange.servicesuitecloud.com` to the Vercel project. At the DNS host for `servicesuitecloud.com`, create the record Vercel shows. That is normally a CNAME from `interchange` to `cname.vercel-dns.com`.

**SEND** "DNS added"

## E Commercial and legal

### E1 Add these to the Ambale thread

In addition to the rate card, could you please confirm in writing the following:

- (a) Whether Metropol reports may be retrieved on behalf of other CBK-regulated lenders through a technology intermediary (the Interchange), under Micromart's subscription. If not, whether each lender would need its own Metropol subscription, with the Interchange acting only as their integration layer.
- (b) Whether we may name Metropol as the source on documents we generate, and refer to Metro-Score and PPI by name.
- (c) Whether our own API documentation may list the code values your API returns (delinquency, product type, account status and response codes) with our own descriptions, without reproducing your guide.
- (d) What is needed to enable report type 22 (Accounts Info) on our account.
- (e) Why the portal PDF for one ID on 15 September listed 50 accounts while API report 8 for the same ID, the same afternoon, returned 48.

**SEND** Metropol's reply when it comes. Until (a) is answered, Bureau Direct is on for Micromart only.

### E2 Axe on paper

You told me to promote Axe to live, and I will. The governance record needs the name of the person deciding and one sentence of reasoning. A signed data-sharing and query agreement with Axe should sit behind it.

**SEND** Decider's name and the rationale sentence

### E3 Counsel

Ask counsel to review two drafts: the Interchange notice in section 5, and a short privacy notice for statement uploads, which I will draft.

**SEND** Counsel's comments

### E4 Prices

Confirm the Interchange fee per product. The current figures are placeholders. Also price the new product, Statement Crunch. My suggestion is free for contributing members up to a monthly allowance, then a fee per statement. Enter Metropol's tariff when it arrives.

**SEND** The fee table

## F People and test data

### F1 Who gets a login

For each person: name, member, role (member admin, analyst or auditor), and email or phone. Also say who holds super admin.

**SEND** The list

### F2 A real M-PESA statement

Send one statement PDF and its password, from someone who agrees to it being used (yours is ideal). If you have an older statement whose password is an ID number, send that too.

**SEND** Save them in `reports\crb\`, which is already git-ignored, and send the passwords in a separate message

### F3 Billed pulls for the preview

This is approval for one billed pull of each of reports 5, 10, 13 and 14 on 30058967, captured word for word and then anonymised. Add 1, 2 and 6 if you want them as their own responses instead of sections of 12. Nothing is pulled without a yes.

**SEND** "Pull 4", "pull 7" or "no pulls"

### F4 The demo date

It is still not written down anywhere, and every "by demo day" depends on it.

**SEND** The date

# My side, in build order

NOW Needs nothing from you

| #  | WORK                              | DONE MEANS                                                                                                                                                                                                        |
|----|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M1 | Report shell v2                   | Letterhead slots (the logo drops in when A1 lands), both notices, request row, provenance lines, sector table, full account appendix, verification QR and footer. The PDF fonts are checked in the rendered file. |
| M2 | Code tables and envelope          | Every Metropol code value typed with our wording, the IX error range, and the Interchange JSON format. The API and the docs read the same tables.                                                                 |
| M3 | Anonymiser, leak scanner, gallery | PDF, JSON, bureau response and CSV for every captured product, and the build fails on any leak.                                                                                                                   |
| M4 | Landing page and docs             | The network landing page on real totals, and /docs with quickstart, reference, code tables, error handbook and snippets.                                                                                          |
| M5 | Port the theatre and engine       | CrunchTheatre, result cards and lib/statement running in the Interchange, with a demo mode on labelled sample data.                                                                                               |
| M6 | Deploy the preview                | Live on Vercel once D1 is done, and on the domain once D2 is done. Every page checked at desktop width and phone width.                                                                                           |
| M7 | Promote Axe                       | AXE-3003 and AXE-3004 set to ACTIVE, recorded as a governance action with your rationale (E2).                                                                                                                    |

AFTER B, C, D The live engine

| #   | WORK                     | DONE MEANS                                                                                                                                                                                                                                                   |
|-----|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| M8  | Move the Registry        | Tables created on the new database. Members, keys, operators, consents, audit and the message log copied, with the chain re-verified after the copy. The four books re-published from live SQL, and p95 measured against 400 ms with every member answering. |
| M9  | Demo borrower candidates | Your answer to brief action 6. Borrowers held by two or more members, clean at one, found from published tokens and resolved to real people only inside the member's own book.                                                                               |
| M10 | Lender portal live       | Sign-in, enquiry workspace, Bureau Direct (Micromart only until E1a), Interchange reports, crunch with metering, the exposure radar, and history with downloads.                                                                                             |
| M11 | Admin console            | Usage events and monthly totals, reciprocity score, member configuration, keys, bureau switch, invoices, audit export.                                                                                                                                       |
| M12 | Hardening                | App role without BYPASSRLS proven by test, passkeys, security headers, rate limits, backups, the verification page, and a penetration-test checklist.                                                                                                        |
| M13 | The specified reports    | 11, then 3, 23, 22 and 21, each with its own acceptance checks.                                                                                                                                                                                              |

WHAT TO EXPECT BY THE END OF THIS SESSION

M1 to M7 done and the preview live on Vercel. The engine move (M8) starts as soon as B, C and D are done, and it can overlap with this session if they arrive early. The full live portal with admin console (M10 to M12) is more than one session of work. I will not mark any of it done until it has run against the real Registry.

A note on M7 and M9: they read and write production (the Registry and the member books). One read-only probe was already blocked by this session's permission layer today. If these are blocked too, I will stop and give you the exact command instead of finding another way in.

## Images, video and effects

Only the logo is required. Most of the motion is built in code, so there is nothing to download for it. Where a stand-in already exists in the Micro Eazy repo, the table says so.

| #  | ASSET                                       | USED ON                                     | SPEC                                                                                                                                        | WHERE TO GET IT                                                                             | NEED          |
|----|---------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|---------------|
| 1  | <b>Logo pack</b>                            | Everywhere                                  | As in A1                                                                                                                                    | Your designer                                                                               | REQUIRED      |
| 2  | <b>Hero video loop</b><br>hero-loop.mp4     | Landing, behind the network                 | 12 to 20 s seamless loop, 1920×1080, no audio, MP4 H.264 under 6 MB, plus a poster JPG. Nairobi at night: light trails, skyline, or aerial. | Pexels Videos or Coverr (free licence). Search "Nairobi night", "city light trails aerial". | OPTIONAL      |
| 3  | <b>At the counter</b><br>counter.jpg        | Landing, the lender band                    | At least 2400 px wide, landscape. A loan officer and a customer at a real Kenyan counter.                                                   | Best: photos at a Micromart branch. Otherwise Unsplash or Pexels, "Kenya bank teller".      | RECOMMENDED   |
| 4  | <b>The borrower</b><br>borrower.jpg         | Landing, "the person behind the token"      | At least 2400 px, daylight: boda rider, market trader, mama mboga                                                                           | Stand-ins already in the app: market-warm, boda-motion                                      | HAVE STAND-IN |
| 5  | <b>Developer band</b><br>docs.jpg           | /docs header                                | At least 2400 px, dark and abstract: circuitry, or a person coding at night                                                                 | Unsplash, "code dark"                                                                       | OPTIONAL      |
| 6  | <b>Member logos</b>                         | Directory, admin, trust strip               | Transparent PNG or SVG                                                                                                                      | Already in micro-eazy-app/public/lenders                                                    | HAVE          |
| 7  | <b>M-PESA backdrop and Safaricom loader</b> | Crunch theatre                              | mpesa-background.jpg, safaricom-25.gif                                                                                                      | Already in the app. Depends on A4.                                                          | HAVE          |
| 8  | <b>Statement how-to film</b>                | Crunch upload screen                        | YouTube Q2Dc036KGnM                                                                                                                         | Already used by the app and the console                                                     | HAVE          |
| 9  | <b>Metropol logo</b>                        | Not used                                    | None                                                                                                                                        | Only with Metropol's written permission (E1b)                                               | NO            |
| 10 | <b>Share image and icons</b>                | WhatsApp and LinkedIn previews, browser tab | 1200×630, favicon set                                                                                                                       | I make these from the logo                                                                  | MINE          |

### Built in code: nothing to download

- **Network in motion** (landing page). Member nodes and query pulses drawn from real totals, never made-up activity.
- **The vault gate**. The cinematic sign-in, which already exists.
- **Exposure radar**. A sweep where each node lights up with its real latency and silent nodes show their reason.
- **Crunch theatre**. Decrypt, parse, extract, ledgers, audit, score.
- **Report typesetting**. Sections assemble onto a page, then the format cards flip in.
- **Product flip cards**. The front asks the question the product answers; the back shows source, price, formats and state.
- **Charts that draw themselves in**. Score dial, PPI ladder, trend line, arrears ladder.
- **Liquid-metal buttons and glow menu**, already in the repo root.
- **Command bar** (Ctrl+K). Jump to an ID, a report or a member from anywhere.
- Every animation has a reduced-motion version and a still state for PDFs and screenshots.

## What is ported from Micro Eazy

| FROM                                         | COMPONENTS                                                                                                                | CHANGE FOR LENDERS                                          |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| micro-eazy-app/components/crunch/Theatre.tsx | MpesaStage, SafaricomLoader, Glass, ExtractStage, ClassifyStage, AuditStage, ScoreDial, Theatre                           | Officer wording, and names on the enquiry instead of "you"  |
| micro-eazy-app/screens/crunch/Cruncher.tsx   | Upload form with the password hint, NameMismatch, ScorePane, LimitPane                                                    | LimitPane becomes an affordability pane. Nothing is booked. |
| connected-suite/components/statement/        | CrunchTheatre (the full-screen counter version, 825 lines), StatementHowTo                                                | The better fit for a lender at a desk, so it is the base    |
| connected-suite/components/                  | RiskBandCard, InternalReportCard                                                                                          | Styled to match the Interchange                             |
| micro-eazy-app/components/                   | Artwork (its generated artwork until a real photo lands), Film, ProgressiveImage, LiquidButton, GlowNav, Deck, AdviceCard | The Interchange palette                                     |
| connected-suite/lib/statement/               | extract-pdf, mpesa-parser, analyze, features, scorecard, thin-file model                                                  | Writes a usage event and keeps no file                      |

## Risks and open decisions

| # | RISK                                                          | WHAT WE DO                                                                                                 |
|---|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 1 | <b>Selling bureau data without Metropol's written consent</b> | Bureau Direct stays Micromart-only until E1a is answered. This is the largest risk in the plan.            |
| 2 | <b>Superuser password exposed on the host</b>                 | Rotate it, take it out of the repo and restrict the port (section 7). Coordinate with CallBox.             |
| 3 | <b>Load on a shared machine</b>                               | Connection limits, timeouts, a memory ceiling and off-peak ingest. Watch the CPU during the first publish. |
| 4 | <b>Safaricom marks in a paid product</b>                      | Your decision on A4                                                                                        |
| 5 | <b>OPRF key in an .env file</b>                               | KMS or HSM before the next member joins (brief action 8)                                                   |
| 6 | <b>Consent minted by the console</b>                          | The Consent Center in PWA onboarding, and a drop-in component for members (brief action 7)                 |
| 7 | <b>Metropol's PDF and API disagree</b>                        | Every figure states its source. E1e asks Metropol why.                                                     |